

ORIGINAL ARTICLE

Automated External Defibrillator Availability, Accessibility, and Usability in High-Risk Out-of-Hospital Cardiac Arrest Neighborhoods of Birmingham, Alabama

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BACKGROUND: Automated external defibrillators (AEDs) are rarely used by bystanders in Birmingham, Alabama. We sought to characterize AED availability, accessibility, and usability in the highest-risk out-of-hospital cardiac arrest neighborhoods of Birmingham.

METHODS: We conducted a descriptive observational study from June 12 to August 27, 2025, in 3 Birmingham neighborhoods with the highest relative out-of-hospital cardiac arrest risk. Potential public AED sites, defined as properties other than single-family residences, were identified using Google Maps landmark descriptions. Sites were contacted by telephone, email, or in person to confirm AED presence, and device-specific data were collected in person. Outcomes included AED availability (presence of ≥ 1 AED), accessibility (public access and hours), and usability (ability to power on with pads). At-risk devices contained expired batteries and pads. Observed and effective AED densities were compared with the recommended 5 AEDs per square mile. Observed AED density was defined as AEDs per square mile, regardless of accessibility hours. Effective AED density was defined as AEDs per square mile adjusted for weekly hours of accessibility.

RESULTS: A total of 287 potential AED locations were identified, and 284 (99%) were contacted and provided data. Fifty-six AEDs were reported at 43 locations (15.1%). Of these, 19 (33.9%) were not publicly accessible. Among the 37 publicly accessible AEDs, none were available 24 hours per day. Thirty-five (94.6%) were found in usable condition, and 5 (13.5%) were classified as at risk. Two (5.4%) were inoperable. The observed density of publicly usable AEDs was 13.5 AEDs per square mile, with an effective density of 3.7 AEDs per square mile due to limited hours of access.

CONCLUSIONS: Despite exceeding the recommended AED density, limited public accessibility and maintenance issues substantially reduce effective AED coverage in Birmingham. Reporting effective AED density, which accounts for hours of access, reveals gaps between device presence and real-world accessibility.

Key Words: adult ■ benchmarking ■ defibrillator ■ hospital ■ telephone

Out-of-hospital cardiac arrest (OHCA) is a leading cause of morbidity and mortality in the United States.^{1–4} Outcomes for OHCA vary significantly across US communities, with overall survival rates ranging from 3.6% to 21.7% in 2024.⁵ Birmingham, Alabama,

a city in the Deep South with a population of $\approx 197\,000$ (67% Black; 24% living below the federal poverty level),⁶ has consistently reported one of the lowest OHCA survival rates in the United States for >15 years.^{7,8} A recent analysis found an overall OHCA survival rate in

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WHAT IS KNOWN

- Survival from out-of-hospital cardiac arrest is strongly influenced by early bystander interventions, including the use of automated external defibrillators (AEDs).
- Birmingham, Alabama, a city in the Deep South, has poor out-of-hospital cardiac arrest outcomes and low rates of bystander AED use.
- AED availability varies by neighborhood characteristics, with devices disproportionately concentrated in higher-socioeconomic areas.

WHAT THE STUDY ADDS

- AED density in 3 high-risk, socioeconomically disadvantaged Birmingham neighborhoods exceeded benchmarks, suggesting low utilization reflects deployment barriers (education, limited access, restricted hours, and maintenance issues) rather than insufficient availability.
- Measurement of effective AED density, which accounts for accessibility over time, reveals a marked gap between nominal device presence and real-world accessibility.

Nonstandard Abbreviations and Acronyms

AED	automated external defibrillator
OHCA	out-of-hospital cardiac arrest

Birmingham of 4.6%, with only 1.3% surviving with a favorable neurological outcome.⁹

These poor OHCA outcomes are likely driven by low rates of OHCA recognition and layperson interventions, including cardiopulmonary resuscitation and automated external defibrillator (AED) use. For example, the cardiopulmonary resuscitation rate in Birmingham is only 16% (2.6× below the national rate),⁸ and a recent 3-year citywide analysis of adult OHCA events identified just 1 case in which a lay community member applied an AED before emergency medical services arrival.⁹ The underlying reasons for the lack of AED utilization are unclear and may involve multiple factors, including the possibility of AED deserts, where devices are not available or publicly accessible in neighborhoods where OHCA occurs. Birmingham currently lacks an AED registry, leaving the number and operational status of publicly accessible devices unknown.

To address this knowledge gap, we conducted an observational descriptive study of AED availability in the 3 highest-risk OHCA neighborhoods in Birmingham, Alabama, recently identified by geospatial analysis of OHCA events and census tract-level data.⁹ High-risk neighborhoods were defined as census tracts within

statistically significant OHCA hot spots (Getis-Ord G_i^* at 99% confidence) that also had social vulnerability index values above the 90th percentile and the highest relative risk of OHCA.⁹ We also sought to evaluate whether the AEDs available were publicly accessible and in usable condition. We hypothesized that these high-risk neighborhoods would represent AED deserts, characterized by few, if any, AEDs, and that available devices would frequently be inaccessible to the public and in unusable condition. The overall goal was to inform targeted community initiatives aimed at increasing bystander response to OHCA, including AED utilization before emergency medical services arrival.

METHODS

Study Aim, Design, and Target Population

We aimed to identify AEDs within select neighborhoods in Birmingham, Alabama, and detail the availability, accessibility, and usability of the individual devices. To do so, we performed an observational descriptive study of AEDs within the 3 highest-risk OHCA neighborhoods in Birmingham, identified in a recent publication.⁹ The University of Alabama at Birmingham institutional review board determined the study to be not human subjects research. The data set is available to replicate the findings of this study on approval by the corresponding author.

The 3 highest-risk OHCA neighborhoods were identified by geospatial analysis of OHCA events in the Birmingham Cardiac Arrest Registry to Enhance Survival from 2020 to 2023.⁹ In brief, ArcGIS Pro (Esri, Redlands, CA) was used to identify neighborhoods within a Getis-Ord G_i^* hotspot that also had an excess relative risk in the top quintile, defined as the ratio of the census tract's observed OHCA rate to the average rate across all tracts.⁹

The social vulnerability index was developed by the Centers for Disease Control and Prevention to evaluate community-level vulnerability. It is derived from the American Community Survey census data and incorporates 4 domains: socioeconomic status, household composition and disability, minority status, and housing and transportation characteristics.¹⁰ These components are combined into a single composite score, with higher values indicating greater social vulnerability. A higher social vulnerability index has been associated with increased age-adjusted cardiac arrest mortality across the United States.¹¹

Analyses were conducted at the census tract level, serving as a proxy for neighborhood. In the prior geospatial analysis, 10 high-risk tracts were identified among 115 total tracts.⁹ The 3 tracts with the highest excess relative risk values were selected from this group as the target neighborhoods for the present study.

Identification of Potential AED Locations With Target Neighborhoods

Potential AED locations were defined as all properties other than single-family dwellings. Examples included parks, recreation centers, apartment complexes, schools, churches, restaurants, banks, private businesses, community health centers, etc.

A GeoJSON file containing the neighborhood boundaries for all 99 neighborhoods within the City of Birmingham, Alabama, was obtained from the city's website and converted into a KML file using Google Earth.¹² This file allowed identification of the official geographic boundaries for all neighborhoods, including the 3 target neighborhoods.

We then cross-referenced each property by street-level block with Google Maps. Property labels from Google Maps were then added to the Google Earth file and verified using embedded satellite imagery and Street View. All properties within the target neighborhoods' geographic boundaries were documented, including name, address, phone number, and property type, and then again manually verified.

Variables/Data Collection

Initial contact of each potential AED location was attempted by telephone or publicly listed email. When successful, a site representative (typically the property manager or owner) was provided a brief explanation of the study and asked if an AED was present on the property. If present, arrangements were made for the study team to visit the AED location to collect device-specific information.

Locations that reported having an AED, as well as those that could not be reached by telephone or email, were visited in person. At each site where an AED was present, the device was examined and a Google Form was completed to document key details, including: the number of AEDs at the site; the exact location of the AED; how the AED was stored; whether it was accessible to the public; the timeframe of accessibility (days of the week and hours of accessibility); the AED model and manufacturer; production date; battery expiration date; whether the device powered on; types of pads available and their expiration dates; and device maintenance arrangements. Data collection occurred over 11 weeks (June 12 to August 27, 2025).

Outcome Measures

The primary outcomes were AED availability, accessibility, and usability. Availability was defined as the number of AEDs located within the target neighborhoods. Accessibility referred to whether AEDs in these neighborhoods were publicly accessible and the days/hours during which they could be accessed. Usability was determined by the device's ability to power on and the presence of electrode pads. Usable devices were further subcategorized as at-risk if the battery was expired, but the device could still power on, or if the available pads were expired.

Secondary outcomes included both observed and effective AED densities for each of the 3 neighborhoods as well as for their combined area. The observed AED density was defined by the number of AEDs per square mile, unadjusted for hours of access. Limited hours of access for existing devices were anticipated; therefore, we also calculated effective AED density by adjusting the observed AED density according to the proportion of weekly hours that AEDs were publicly accessible. This metric, as calculated using the equation in Figure 1, included only usable devices. To our knowledge, effective AED density has not been previously described and represents a novel approach to incorporating temporal accessibility into AED density estimates. However, this measure has limitations, including the assumption that accessibility is constant within

$$\text{Effective AED Density} = (\text{AEDs per square mile}) \times \left(\frac{\text{Mean hours accessible per week}}{168} \right)$$

Figure 1. Effective automated external defibrillator (AED) density.

Equation used to adjust observed AED density based on the proportion of weekly hours that devices are publicly accessible.

reported hours, potential misclassification of access times, and the inability to account for real-time barriers to access.

The area in square miles for each target neighborhood was obtained from the 2023 US Census Bureau American Community Survey.¹³ The data from the survey are at the census tract level, which was used as a proxy for neighborhoods, consistent with previous reports.^{14–16} The benchmark AED density was set at 5.2 AEDs per square mile (or 2 AEDs per square kilometer), based on the findings of Stieglis et al¹⁷ All data were summarized using descriptive statistics.

RESULTS

The 3 target neighborhoods comprised a total population of 3254 individuals across 2.6 square miles, resulting in a population density of 1252 persons per square mile. The majority of residents were female (60.2%) and identified as Black (87.8%), and nearly half of the population (48.7%) lived below the federal poverty level. Additional demographic and socioeconomic characteristics are summarized in Table 1.

Across the 3 neighborhoods, 287 potential AED locations were identified, and 284 (99%) were successfully contacted. Among the contacted sites, 241 (84.9%) reported that they did not have an AED on the premises. The remaining 43 locations (15.1% of contacted sites) reported 56 available AEDs (Table 2). These AEDs were distributed across a range of site types, including private businesses (34 devices, 60.7%), public schools (11 devices, 19.6%), parks and community centers (4 devices, 7.1%), churches (4 devices, 7.1%), and other locations (3 devices, 5.4%).

Of the 56 available AEDs, 19 (33.9%) were not accessible to the public (Figure 2); all were in private businesses that reported the devices were not for public use. The remaining 37 (66.1%) were reported to be publicly accessible. Thirty-five devices (94.6%) were found in usable condition, while 2 (5.4%) were inoperable due to a dead battery (1 device located in a private business) or missing pads (1 device located in a community center). Additionally, 5 usable devices (13.5%) were subclassified as at risk because of expired batteries and pads, although they retained the ability to power on.

Among the 35 devices in usable condition, 19 (54.3%) had pediatric-designed pads, including 3 devices equipped with manufacturer-designated combination pads suitable for both adults and children. Sixteen devices (45.7%) had no pediatric-specific pads available, leaving only adult pads for use. Of the 16 devices with dedicated pediatric pads (ie, excluding combination

Table 1. Demographic and socioeconomic characteristics of study neighborhoods in Birmingham, Alabama

Metric	Neighborhood A	Neighborhood B	Neighborhood C	Total
Population, n	1028	1365	861	3254
Median age, y	41.4	13.8†	30.9	n/a*
Female, %	57	59	66	60.2
Black race, %	94	85	83	87.8
Median Household income, \$ USD	30 841	32 972	30 222	n/a*
Population living below poverty line, %	32.7	61.3	45.4	48.7
High school graduation or higher, %	83.2	87.4	84.2	85.1
Bachelor's degree or higher, %	8.3	13	3.6	9.1
Area, sq mile	0.4	1.9	0.3	2.6
Population density, persons per sq mile	2299	708	3000	1252

*Unable to aggregate median values due to lack of individual-level census data.

†60% of the neighborhood B population is under 18 y of age, with 32% below 9 y of age.

pad devices), 3 (18.8%) had expired pads (all expired in 2022). A total of 5 devices (14.3%) had expired adult pads.

The observed AED densities within each neighborhood exceeded the benchmark of 5.2 AEDs per square mile (Figure 3). The density of available devices ranged from 12.5 to 30 AEDs per square mile, with an overall observed density of 21.5 AEDs per square mile. The density of publicly accessible devices ranged from 12.5 to 26.7 AEDs per square mile, with an overall density of 14.2 AEDs per square mile. The density of usable AEDs ranged from 10 to 26.7 AEDs per square mile, resulting in an overall density of 13.5 AEDs per square mile, which is 2.5× the benchmark density.

Among the publicly accessible AEDs, none were available 24 hours per day. No devices were accessible between 23:00 and 06:00, and only 27% were available at any time on weekends. The mean (SD) hours of AED accessibility at locations with usable devices were 46.6 (21.3) hours per week overall, 36.3 (21.4) in neighborhood A, 48.7 (18.3) in neighborhood B, and 46.2 (30.2) in neighborhood C (Table S1). When adjusted for

mean weekly hours of access, the effective density of usable AEDs decreased to a range of 2.2 to 7.3 AEDs per square mile, with an overall effective density of 3.7 usable AEDs per square mile, below the benchmark density.



DISCUSSION

This descriptive observational analysis of AEDs in the 3 highest-risk OHCA neighborhoods in Birmingham, Alabama, found that 1 in 3 AEDs were inaccessible to the public and that 1 in 5 publicly accessible AEDs were either inoperable or at risk of failure due to expired batteries or electrode pads. Lastly, although the observed density of usable devices exceeded the recommended benchmark in each neighborhood, the effective density was reduced >3-fold because of limited hours of access, with no AEDs available to the public 24 hours per day.

The lack of public access to one third of AEDs was a surprising finding that warrants additional context. Site representatives at locations reporting 1 or more AEDs

Table 2. Availability, Accessibility, and Usability of Automated External Defibrillators by Neighborhood

Metric	Neighborhood A	Neighborhood B	Neighborhood C	Total
Available AEDs, n	5	42	9	56
Density (AEDs per sq mile)	12.5	22.1	30	21.5
AEDs per 1000 population	4.9	30.8	10.5	17.2
Accessible AEDs, n (%)*	5 (100)	24 (57.1)	8 (88.9)	37 (66.1)
Density (AEDs per sq mile)	12.5	12.6	26.7	14.2
AEDs per 1000 population	4.9	17.6	9.3	11.4
Usable AEDs, n (%)†	4 (80)	23 (95.8)	8 (100)	35 (94.6)
Density (AEDs per sq mile)	10	12.1	26.7	13.5
AEDs per 1000 population	3.9	16.9	9.3	10.8

AED indicates an automated external defibrillator.

*Denominator=available AEDs.

†Denominator=accessible AEDs.

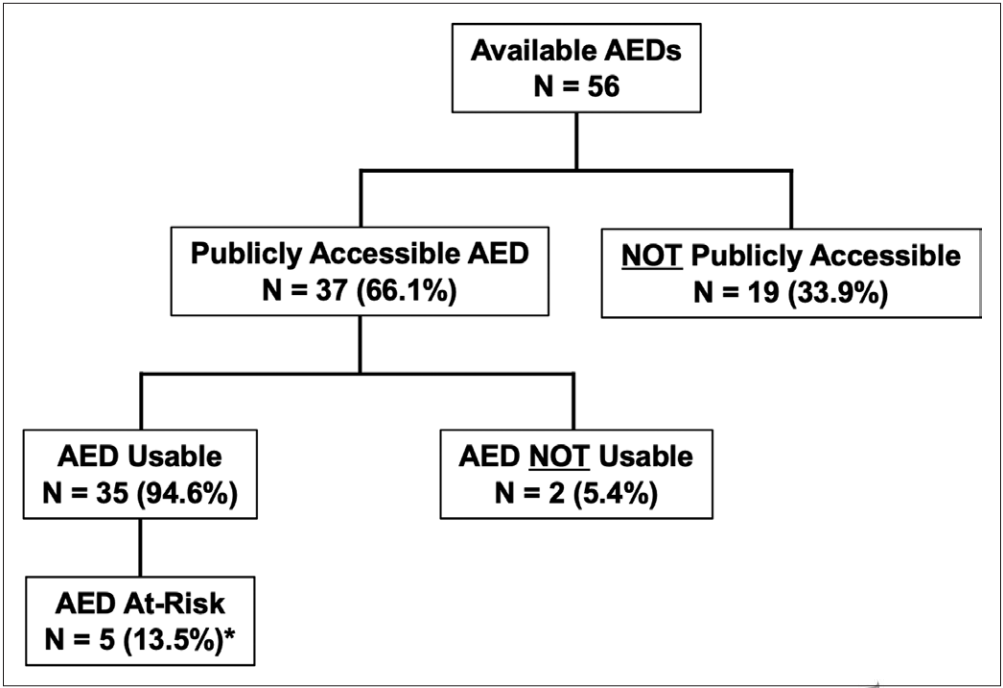


Figure 2. Flow diagram of automated external defibrillator (AED) classification.
*Denominator=37 publicly accessible AEDs.



were presented with the following scenario to determine public accessibility: “If a member of the public suffered a cardiac arrest on the property or across the street, could the onsite AED(s) be used for their treatment?”

Many private businesses responded no, explaining that their AEDs were intended for employee use only and not for the general public. Motivations behind these internal policies were not explored but may be due to gaps in

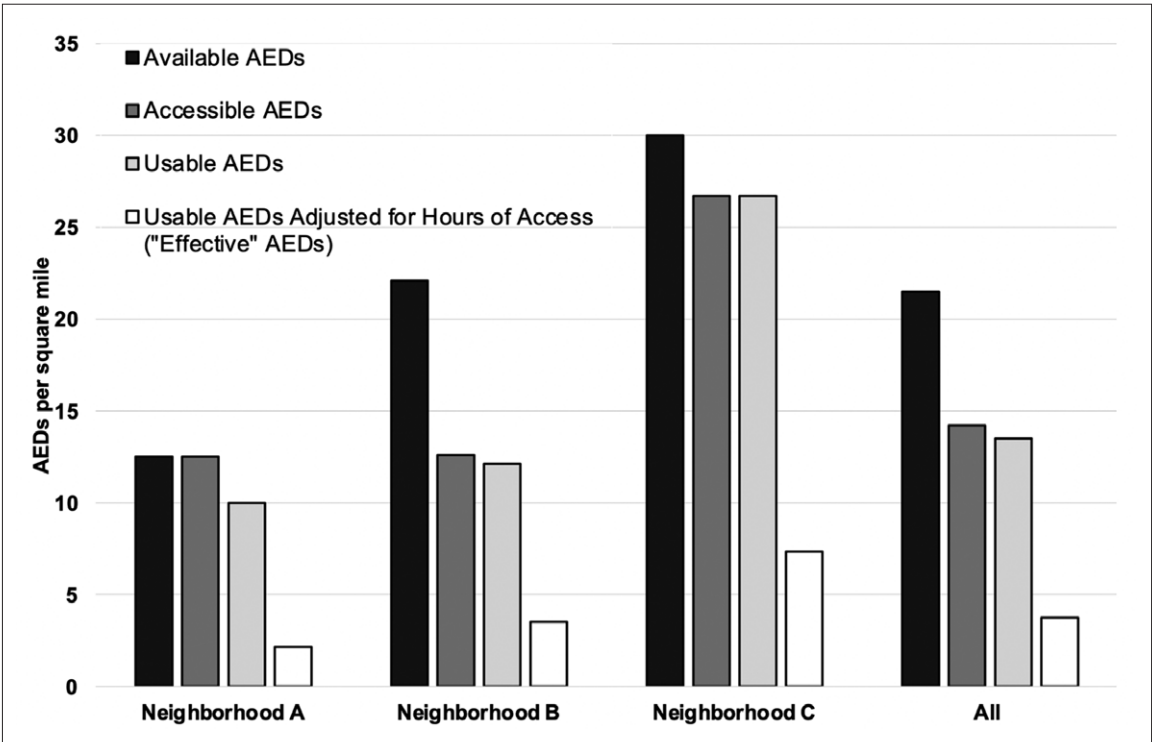


Figure 3. Automated external defibrillator (AED) densities stratified by device availability, accessibility, and usability classifications.
Benchmark density=5.2 AEDs per sq mile.

awareness regarding the broader public health role of AEDs or uncertainty about legal protections under Good Samaritan laws. Because workplace OHCA are classified as occurring in a public location within surveillance registries such as Cardiac Arrest Registry to Enhance Survival,¹⁷ some of these nonaccessible AEDs are technically available for treatment of certain public OHCA (ie, employees) but reportedly unavailable for bystander use in nearby incidents.

In addition to accessibility, device maintenance is a well-recognized barrier to successful AED deployment.¹⁸ Expired batteries can impair a device's ability to analyze cardiac rhythm and deliver an effective shock, while expired electrode pads may dry out, limiting chest adhesion and shock conduction. In our analysis, nearly one in 5 AEDs contained expired batteries or pads or were unable to power on. This finding is consistent with the work of Jespersen et al,¹⁹ who reported a 20% rate of nonfunctional AEDs (including those with expired batteries and pads) in Bornholm, Denmark. Unlike our analysis, their work was limited to registered AEDs, underscoring that even high-performing systems with established, robust infrastructure (ie, municipal AED registry) remain vulnerable to lapses in device maintenance. AED registries are uncommon in the United States, suggesting that similar maintenance vulnerabilities are likely present in other communities.

A particular maintenance concern identified was related to access to pediatric-specific electrode pads. Nearly half of the devices lacked pediatric pads, leaving only adult pads available for use. Guidelines support the use of adult pads for children if no pediatric pads are available,²⁰ thus these devices remain technically usable. However, the absence of pediatric pads for noncombination pad devices may introduce unnecessary hesitation among lay responders. Furthermore, among devices equipped with pediatric pads, approximately 1 in 5 had expired pads, potentially compromising optimal shock conduction and highlighting an additional layer of maintenance vulnerability.

Our between-neighborhood findings are likely influenced by differences in neighborhood structure and composition. For example, neighborhood B contained the largest number of available AEDs (n=42) but was spread across a much larger geographic area and characterized by a substantial commercial presence with relatively low residential density. In contrast, neighborhood C had fewer AEDs overall (n=9) but a higher AED density, driven by its much smaller geographic area and more concentrated population. Despite this higher spatial density, neighborhood C had fewer AEDs per population than neighborhood B (9.3 versus 16.9 per 1000 persons), underscoring the distinction between population-based and spatial-based availability of AEDs. This is consistent with prior work from England demonstrating that AEDs are disproportionately located in more

affluent neighborhoods and areas with higher workplace density, highlighting a broader association between AED availability and community socioeconomic characteristics.²¹ Future work is needed to better characterize the structural makeup of the neighborhoods included in our analysis and to evaluate how block-level population density and land-use patterns influence AED availability and accessibility.

Our findings have important implications for public policy and local-level planning. Alabama currently has limited state statutes and regulatory provisions governing AEDs. Existing statutes require AED placement in all public K-12 schools,²² dialysis centers,²³ and assisted living facilities.²⁴ Owners of AEDs are encouraged to complete a nationally recognized cardiopulmonary resuscitation/AED training course, and devices are required to be maintained and tested in accordance with manufacturer guidelines.²⁵ However, there are no regulations requiring AED registration or notification following device use. In the 3 neighborhoods evaluated, 2 public K-12 schools were identified, accounting for 11 of 56 devices (20%), and no dialysis centers or assisted living facilities were present, underscoring the limited representation of state-mandated AED locations within the study area.

Ideally, AED locations would be registered and integrated with 9-1-1 dispatch systems to enable telecommunications to direct callers to the nearest available device. Implementing such a system at scale would likely require an approach similar to the methodology used in the present analysis. Although resource-intensive and requiring 11 weeks to complete assessments in 3 of the city's 99 neighborhoods, this approach ensured consistent, high-quality data collection. Community-sourced AED identification through mobile applications,^{26,27} may represent a complementary or alternative strategy to support broader AED mapping efforts but may sacrifice some level of data quality.

Temporal accessibility of public AEDs is another concerning finding. A recent special report from the *American Heart Association Automated External Defibrillator Symposium* recommended AED placement in locations with 24/7 accessibility.²⁸ However, none of the devices identified in our analysis were available around the clock, with access largely restricted to weekday business hours. This lack of 24/7 access to public AEDs has been noted in other locations including Copenhagen, Denmark²⁹ and Seoul, South Korea.³⁰ To address this gap, we introduced an additional metric, effective AED density, which adjusts observed AED density by the mean weekly hours of access and demonstrated a reduction of more than 3-fold under current accessibility conditions in Birmingham. This measure may be useful for other communities that rely on AED density measures to evaluate their public access defibrillation programs. The limited hours of accessibility in Birmingham substantially diminish the potential impact of existing devices. For example, on

review of recent OHCA in Birmingham, $\approx 48\%$ occurred outside typical business hours (08:00–17:00).

Our previous geospatial analysis of the Birmingham Cardiac Arrest Registry to Enhance Survival data, which identified the at-risk neighborhoods used in the present study, found only 1 OHCA event in which a lay community member applied an AED before emergency medical services arrival.⁹ Based on this finding, we initially hypothesized that a lack of public AEDs, resulting in relative AED deserts in high-risk locations, contributed to the low utilization. However, our present findings suggest that low AED use may be more strongly related to restricted hours of access, as described above, and to limited community awareness and education regarding the role of AEDs and municipal strategies for their deployment. Targeted educational efforts, along with consistent AED signage and visible placement, could help address this gap. For example, during a site visit at a church, 1 pastor retrieved an AED from a closet in his office and handed our study team the device housed in an unmounted cabinet, which was intended to be wall-mounted for public visibility. The device had been donated to the church a year prior. This anecdote underscores that merely increasing the volume of AEDs without accompanying implementation strategies is unlikely to be effective.

Limitations

There are several limitations to this study that should be considered. First, most sites were initially contacted by telephone to inquire about AED availability. This approach may bias results toward underreporting if site representatives inaccurately reported that no AEDs were available. However, sites reporting publicly accessible AEDs were subsequently confirmed through in-person site visits, reducing the risk of overreporting.

Second, the results reflect data collected during an 11-week assessment period. Device status may have changed since data collection, including maintenance of existing devices, placement of new AEDs, or expansion of access and availability. Ongoing surveillance will be necessary to monitor changes over time.

The AED density benchmark of 5.2 AEDs per square mile was derived from the work of Stieglis et al and is based on European data.¹⁷ There is currently no specific recommendation for AED density in the United States; instead, the American Heart Association HeartSaver program recommends placing AEDs within a 3- to 5-minute round-trip walk from potential OHCA locations.³¹ Future work is needed to evaluate spatiotemporal measures of AED accessibility in Birmingham.^{32,33}

Finally, the 3 high-risk neighborhoods included in this analysis were identified using the findings of our previously published geospatial analysis of all adult nontraumatic OHCA events, excluding emergency medical services-witnessed events.⁹ This approach did not

account for arrest location type and therefore included events occurring in private residences. Future studies should evaluate neighborhoods at highest risk for public-location OHCA specifically, which may differ from the neighborhoods assessed in the present analysis.

Conclusions

This descriptive observational study of 3 high-risk OHCA neighborhoods in Birmingham, Alabama, found that although AED density exceeded recommended benchmarks, substantial barriers to effective AED deployment exist. One-third of AEDs were not publicly accessible, 1 in 5 accessible AEDs had maintenance-related deficiencies, and limited hours of public accessibility reduced the potential impact of the present devices. These findings underscore the need for a strengthened public access defibrillation program in Birmingham that prioritizes improved accessibility, routine device maintenance, and community education to enhance OHCA outcomes.



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Author Contributions

R.A. Coute designed the study, acquired the funding, performed data collection, analyzed the data, participated in article writing, and provided senior supervision. J.M. Smith and Dr Hudson performed data collection, data analysis, and participated in article writing. J.G. Evans, Dr White, Dr Ferguson, Dr Jackson, and Dr Kurz participated in article writing and editing. All authors have read and approved of the article.

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Disclosures

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Supplemental Material

Table S1

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